logical barriers for market entry and come up with unique solutions to meet the needs of their prospective clients.

Yet to commercialise

Liu opines that more applications lead to commercialisation. “The primary market is display technology—ultra large displays. Meanwhile, with other applications demanding a lot of innovations, it may take around 4-5 years for the automotive industry, 2-3 years for the others to get commercial, and help to explore larger possibilities of adopting µLED tech in future.”

He adds that reducing the cost and number of defects while improving yield is the way to commercialise µLEDs. “Our research team has different challenges compared to mass production, yet innovations and trials can be achieved with experience.”

Despite the manufacturing struggles, innovators find µLEDs to be worth pursuing. Along with around 50% reduction in power consumption compared to OLEDs, these displays also offer longer lifespans. µLEDs do not exhibit burn issues, as in the case of OLEDs. Even so, many companies have failed to enter mass production and are far from reaching average consumers. The reduced cost of µLED and mini LED products and components will drive their adoption in the market. For the R&D engineers to achieve true design freedom, the primary problem of mass transfer and fast placement with accuracy has to be addressed first.

In conversation with EFY, Ron Mertens, CEO and founder of Metalgrass, a technology knowledge hub that is focussed on display technologies, says that commercialisation may take some time. “For µLEDs, there are many technical challenges in epitaxy, transfer, inspection/repair, full-colour displays, and more. The cost issue adds to it.”

He estimates that µLED adoption, which is likely to happen within two to five years, would begin mostly in three market segments—large-format TVs, microdisplays (for AR/VR), and perhaps simple wearables and automotive. “I think adopting this technology for applications such as smartphones, IT, TVs will take a longer time in the mass market. Even if these are just predictions, it is too early to call,” he adds.

However, to further materialise the µLED displays, innovators have targeted optimising manufacturing technology and reducing production cost. With an even tinier chip size, µLEDs require enhanced equipment and semiconductor-level equipment to process. Cooperation between the supply chains is pivotal for µLED chip manufacturing, mass transferring, testing, and repairing. Working with semiconductor material and equipment suppliers could be a feasible solution for µLED display production.

Can µLEDs replace existing displays?

“They have the best performance, brightness, concept ratio, and are also reliable and flexible. A tech with immense potential, which is why we think it is disruptive,” says Liu. Indeed, µLEDs are an interesting technology with differentiating features.

While many companies do seem to be aggressively working on imbibing it in their new products, the heavy investments tend to push a few off the track. It might take some years for this emerging tech to become mainstream. Till then, the decision of investment depends on the application, and whether µLEDs can bring a greater benefit than incumbent technologies to the table. EFY

Abbinaya Kuzhanthaivel is an Assistant Editor at EFY